HARD · ADVANCED MATH

Advanced Math

02

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ADVANCED MATH

$$x - 29 = x - ax - 29$$

Which of the following are solutions to the given equation, where a is a constant and $a > 30$?

I. a
II. $a + 1$
III. 29

A. I and II only

B. I and III only

C. II and III only

D. I, II, and III

Q2 ADVANCED MATH

$$\frac{2}{x-2}+\frac{3}{x+5}=\frac{rx+t}{(x-2)(x+5)}$$

The equation above is true for all $x>2$, where r and t are positive constants. What is the value of rt ?

A. -20

B. 15

C. 20

D. 60

Q3

ADVANCED MATH

What is the minimum value of the function f defined by $f(x) = (x-2)^2 - 4$?

A. -4

B. -2

C. 2

D. 4

Q4

GRID-IN ADVANCED MATH

$$x^2 + 7x + 5 = 0$$

One solution to the given equation can be written as $x = \frac{-7 + \sqrt{k}}{2}$, where k is a constant. What is the value of k?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5 GRID-IN ADVANCED MATH

$$\sqrt[5]{70n}\ \sqrt[6]{70n}\ ^{2}$$

For what value of x is the given expression equivalent to $70n^{30x}$, where $n > 1$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6 ADVANCED MATH

$$fx = 5{,}4700.64^{\frac{x}{12}}$$

The function f gives the value, in dollars, of a certain piece of equipment after x months of use. If the value of the equipment decreases each year by $p\%$ of its value the preceding year, what is the value of p?

A. 4

B. 5

C. 36

D. 64

Q7

ADVANCED MATH

$$8x + y = -11$$

$$2x^2 = y + 341$$

The graphs of the equations in the given system of equations intersect at the point (x, y) in the xy-plane. What is a possible value of x?

A. -15

B. -11

C. 2

D. 8

Q8

ADVANCED MATH

Which of the following expressions is equivalent to $\frac{x^2-2x-5}{x-3}$?

A. $x - 5 - \frac{20}{x-3}$

B. $x - 5 - \frac{10}{x-3}$

C. $x + 1 - \frac{8}{x-3}$

D. $x + 1 - \frac{2}{x-3}$

Q9 GRID-IN ADVANCED MATH

x	y
21	-8
23	8
25	-8

The table shows three values of x and their corresponding values of y, where $y = fx + 4$ and f is a quadratic function. What is the y-coordinate of the y-intercept of the graph of $y = fx$ in the xy-plane?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10

ADVANCED MATH

$$-2x^2 + 20x + c = 0$$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c?

A. -68

B. -50

C. -32

D. 0